

Check-In 4: Evaluation Results & High-Fidelity Prototype

Calerb Louis Jean, Haoran Huo, Xinqi Tao

Group Name: HeyBene

cjean33@gatech.edu | hhuo30@gatech.edu | taoxinqi23@gatech.edu

1 INTRODUCTION

In Check-In 3, we outlined our evaluation plan and built three functional web prototypes to test different interaction concepts for teleoperating our robot. As stated in our previous report, the core question we aimed to answer was which approach best bridges the tactile gap while making the current drive mode clear at all times.

This document serves as the direct continuation of that work. We first present the comprehensive results and analysis from our usability testing sessions. Following that, we detail how these specific user insights drove our design iterations, ultimately resulting in our finalized High-Fidelity (Hi-Fi) prototype.

2 EVALUATION RESULTS & ANALYSIS

We conducted moderated, one-on-one sessions with 5 participants using the Think-Aloud Protocol, following the exact procedure outlined in Check-In 3. Our participant pool intentionally included a mix of experienced mobile gamers and users with no prior robotics or RC background.

2.1 Quantitative Findings

Participants rated their experience on a 7-point Likert scale across three dimensions: Ease of Steering, Confidence in Finger Placement, and Clarity of Mode Status. We also tracked task completion regarding mode switching.

- **Alternative 1 (Mobile Gamer):** This layout emerged as the clear quantitative winner. It scored exceptionally high in Ease of Steering (mean: 6.0/7) and Clarity (mean: 6.4/7). Participants completed the mode-switching task instantly.
- **Alternative 2 (Immersive FPV):** This prototype performed poorly across all metrics, particularly in Confidence (mean: 3.4/7) and Ease of Steering (mean: 2.6/7). Users reported that the tilt-to-steer mechanic felt unstable.

- **Alternative 3 (One-Handed Casual):** This approach yielded moderate scores (Ease: 4.3/7, Confidence: 5.3/7). While users appreciated the novelty, it lacked the precision required for navigating the S-curve task.

2.2 Qualitative Insights

By analyzing the think-aloud transcripts and post-test interviews, we synthesized three core themes that address our initial needfinding questions:

- 1. Active Braking vs. Passive Release (The Mental Model Clash):** We initially hypothesized that the "release-to-brake" mechanic in Alternative 3 would provide the safest user experience. However, qualitative feedback strongly contradicted this. P3 and P5 explicitly noted that in an emergency, human muscle memory favors an active action—pressing a dedicated button. Stopping when input ceases is viewed as a baseline default, not a substitute for an active brake. The physical "Brake" button in Alternative 1 satisfied this psychological need for active defense.
- 2. Axis Coupling and Ergonomic Strain:** Testing revealed that combining directional and speed controls into a single gesture severely hinders precision. In Alternative 3, attempting to map both X (turn) and Y (speed) to one thumb caused the robot to move in rigid lines rather than smooth curves. Furthermore, P2 pointed out a severe ergonomic flaw: single-handed swipe patterns are physically straining on large modern smartphones due to a restricted reachability zone. Decoupling these axes (left thumb for direction, right thumb for speed) in Alternative 1 felt highly natural and predictable.
- 3. Visual Stability over Minimalist UI:** Alternative 2 attempted to maximize the FPV camera feed by relying on gyroscope steering and hiding most UI elements. This approach failed for two reasons. First, P4 and P5 reported that physically tilting the phone caused visual fatigue and broke their concentration on the live feed. Second, the hidden UI led to severe mode confusion; participants did not know how to engage Auto Tracking. Users prefer a visible, stable UI (like Alternative 1) even if it slightly obstructs the camera feed, as it provides a higher sense of system control.

3 DESIGN ITERATIONS: FROM EVALUATION TO HI-FI

Based on the clear preference for Alternative 1, we selected the dual-joystick/button layout as our primary interface. We implemented the following critical design iterations in our High-Fidelity prototype to resolve the specific pain points identified:

- **Dedicated Brake Retention:** We solidified the large, capsule-shaped "Brake/Decelerate" button on the right side to ensure users always have an active defense mechanism.
- **Glassmorphism for FPV Visibility:** To address concerns about opaque UI elements blocking the camera feed, we applied semi-transparent overlays (alpha channel adjustments) to the directional joystick and action buttons. This maintains the HUD functionality while preserving background visibility.
- **Dynamic Joystick Anchoring:** To accommodate different hand sizes and solve the reachability issue, the left joystick is now dynamic—it centers exactly where the user's thumb first touches the screen, rather than being fixed in one spot.
- **Stable Information Hierarchy:** The fluctuating numerical battery text (which distracted P2) was replaced with a stable visual progress bar.

4 HIGH-FIDELITY PROTOTYPE IMPLEMENTATION

Our Hi-Fi prototype transforms the validated interaction concepts into a fully styled, responsive frontend application optimized for landscape orientation.

4.1 Interface Architecture

[PLACEHOLDER: Insert screenshot of the new frontend UI here]

The interface is built using HTML5, CSS3, and JavaScript. To ensure the interface feels like a professional controller, we utilized the native Touch Events API. This allows for true multi-touch capability, meaning a user can continuously adjust steering with their left thumb while simultaneously tapping the brake with their right thumb without the inputs interfering.

4.2 Next Steps

With the Hi-Fi frontend interface established, our final phase will focus on hardware integration. We will connect the frontend WebSocket events to the actual OpenBot hardware to test real-world latency, telemetry synchronization, and the responsiveness of the AI Auto-Tracking mode toggle.

A APPENDIX: USABILITY TESTING RAW NOTES

Participant 1 (Profile: Novice / No prior robotics experience)

- **Tested Order:** Alt 1 → Alt 2 → Alt 3
- **Alt 1 (Mobile Gamer):** Noticed battery dropping (95% to 92%). Identified camera feed. Dynamic joystick felt too sensitive without physical resistance; micro-adjustments were hard. Found the Auto-Tracking toggle but didn't fully understand what it did. (Scores: Ease 4, Conf 6, Clarity N/A)
- **Alt 2 (Immersive FPV):** Gyroscope steering was a struggle. Hard to keep the phone steady. Major anxiety due to the lack of an obvious brake/stop mechanism. (Scores: Ease 2, Conf 1, Clarity 6)
- **Alt 3 (One-Handed):** Auto-brake (release to stop) solved the anxiety from Alt 2. Steering was okay but less intuitive than a dual-joystick setup. (Scores: Ease 3, Conf 5, Clarity 5)
- **Final Choice:** Alt 1. Stated clear inputs are the priority for safety.

Participant 2 (Profile: Experienced Gamer / Large Smartphone User)

- **Tested Order:** Alt 2 → Alt 3 → Alt 1
- **Alt 2 (Immersive FPV):** Disliked numerical battery jumping; suggested a visual bar. Gyro steering felt unnatural ("nobody twists their phone around"). UI was too minimal to find the Auto mode. Low trust in AI without visual feedback on the feed. (Scores: Ease 3, Conf 5, Clarity 2)
- **Alt 3 (One-Handed):** Pointed out a severe ergonomic flaw: single-handed swipe is physically straining on a large phone. Reachability zone is too small. Liked the release-to-stop mechanic.
- **Final Choice:** Selected Alt 1 as default. Suggested keeping Alt 2/3 hidden in settings for niche preferences.

Participant 3 (Profile: Precision-focused / Prefers active control mechanics)

- **Tested Order:** Alt 3 → Alt 1 → Alt 2
- **Alt 3 (One-Handed):** Hard to map both X (turn) and Y (speed) on one thumb. Robot moved in rigid lines, couldn't draw smooth curves. Found "release to brake" counter-intuitive; braking should be an active press in an emergency. (Scores: Ease 3, Conf 6, Clarity 4)

- **Alt 1 (Mobile Gamer):** Relieved to have speed and direction decoupled. Predictable mapping. Loved the physical brake button for active stopping. (Scores: Ease 6, Conf 6, Clarity 6)
- **Alt 2 (Immersive FPV):** Tilt steering lacked fine precision. Vertical slider was terrible for emergency braking. (Scores: Ease 3, Conf 2, Clarity 5)
- **Final Choice:** Alt 1. Active braking and clear mapping won out.

Participant 4 (Profile: Experienced Gamer / Familiar with mobile racing games)

- **Tested Order:** Alt 1 → Alt 2 → Alt 3
- **Alt 1 (Mobile Gamer):** Immediately recognized the layout as being similar to "QQ Speed" (QQ). Used system status icons to confirm connectivity. Confirmed that the split controls (left for direction, right for gas) felt natural and highly predictable. (Scores: Ease 7, Conf 5, Clarity 7)
- **Alt 2 (Immersive FPV):** Found tilt-steering on a single smartphone slab uncomfortable and ergonomically restrictive. Tilting caused visual fatigue and hindered concentration on the live feed. (Scores: Ease 1, Conf 6, Clarity 4)
- **Alt 3 (One-Handed):** Acknowledged the novelty and situational utility, but lacked precision for long sessions. Mode banner was too visually distracting. (Scores: Ease 5, Conf 5, Clarity 5)
- **Final Choice:** Strongly favored Alt 1 as the primary interface. Suggested keeping Alt 3 as a "Quick-Access" secondary mode.

Participant 5 (Profile: Logical/Skeptical Minimalist)

- **Tested Order:** Alt 2 → Alt 3 → Alt 1
- **Alt 2 (Immersive FPV):** Conceptual mapping failure: real steering wheels rotate, they don't tilt based on gravity. Expressed frustration with the "shaking" visual experience. (Scores: Ease 4, Conf 3, Clarity 4)
- **Alt 3 (One-Handed):** Argued that stopping upon input cessation should be a default expectation, not a feature. The absence of a physical brake button leads to anxiety and a lack of active control. (Scores: Ease 5, Conf 5, Clarity 5)
- **Alt 1 (Mobile Gamer):** Preferred this for its superior sense of control. Acknowledged the trade-off: buttons slightly obstructed the FPV feed, but feeling in control is more important for a robot. (Scores: Ease 6, Conf 7, Clarity 6)
- **Final Choice:** Chose Alt 1. Prioritized active control and predictable mapping over visual minimalism.